

My First LaTeX Document

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1.1

2.

$$\{3x + 2 : x \in \mathbb{Z}\}$$
$$\{\dots, -4, -1, 2, 5, 8, \dots\}$$

12.

$$\{x \in \mathbb{Z} : |2x| < 5\}$$
$$\{0, 2, 4\}$$

26.

$$\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, \dots\}$$
$$\{3^x : x \in \mathbb{Z}\}$$

34.

$$|\{x \in \mathbb{N} : |x| < 10\}|$$

The set has a cardinality of 10.

1.2

2.

$$\{1, 2, \emptyset\}$$
$$\{1\}, \{2\}, \{\emptyset\}, \{1, 2\}, \{1, \emptyset\}, \{2, \emptyset\}, \{1, 2, \emptyset\}$$

3.

$$\{\{\mathbb{R}\}\}$$
$$\{\mathbb{R}\}$$

4. $\emptyset$

$$\emptyset$$

1.3

- 8.
- 9.

1.4

- 8. $\mathcal{P}(\{1,2\} \times \{3\})$
 $\{3\}, \{6\}, \{3,6\}$

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